

Ibexolimod Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-125/A

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Deviations observed during the campaign were investigated and closed with no impact to product quality. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

The control strategy links critical quality attributes to critical process parameters identified during development. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Section 3.2.P.4 — Excipients

For the commercial formulation F-125/A, Ac-Di-Sol SD-711 (Croscarmellose sodium, disintegrant) is used. Solutab Type A was included in the excipient compatibility screen but was not carried forward into the commercial formulation.

The approach described herein is consistent with IPEC Excipient Qualification Guide and Ph. Eur. general monograph 2034.

Justification of Material Grade

The formulation development rationale for Ac-Di-Sol SD-711 rests on superior compactibility at low compression forces.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Croscarmellose sodium	disintegrant	Ac-Di-Sol SD-711
Reference		SPEC-EX-4685 Rev 09

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

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process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission.